Save My Exams

GCSE · OCR · Computer Science

51 mins 16 questions

Programming Languages & Integrated Development Environments (IDEs)

Programming Languages & Integrated Development Environments (IDEs)

Levels of Programming Languages / Translators, Compilers & Interpreters / Tools & Facilities in IDEs

Easy (5 questions)	/11
Medium (8 questions)	/28
Hard (3 questions)	/12
Total Marks	**/51**

Scan here to return to the course
or visit savemyexams.com

© 2026 Save My Exams, Ltd.

Easy Questions

1 (a) A computer game is written in a high-level programming language.

State why the computer **needs to translate** the code **before it is executed**.

..... **(1 mark)**

(b) Either a **compiler** or an **interpreter** can translate the code.

Describe two differences between how a compiler and an interpreter **would translate the code**.

.....

.....

.....

..... **(4 marks)**

2 **Define** a low-level programming language?

.....

..... **(2 marks)**

3 **State** the **primary advantage** of high-level programming languages?

..... **(1 mark)**

4 **Define** the role of an **assembler**?

..... **(1 mark)**

5 Programs can be written in high-level languages or low-level languages.

Give **two** reasons why some programs are written in a low-level language.

© 2026 Save My Exams, Ltd.

.....

..... **(2 marks)**

Medium Questions

1 **Complete the description** of programming languages and translators by **writing the correct term** from the box in each space.

continues	crashes	debugging	error	executable
high-level	interpreter	language	low-level	many
no	one	stops	with	without

Jack writes his program in a ..... language. This needs to be translated into assembly or machine code before it can be executed. This is done using a translator.

One type of translator is an interpreter. This converts one line of code and then executes it, before moving to the next line. It ..... when an error is found, and when corrected continues running from the same position. This translator is helpful when debugging code.

A second type of translator is a compiler. This converts all of the code and produces an error report. The code will not run until there are ..... errors.

The ..... file produced can be run ..... the compiler.

(5 marks)

2 The area of a circle is calculated using the formula π × r2 where π is equal to 3.142 and r is the radius.

A program is written to allow a user to enter the radius of a circle as a whole number between 1 and 30, then calculate and output the area of the circle.

```
radius = 0

area = 0.0

radius = input("Enter radius")

if radius < 1 OR radius > 30 then

    print("Sorry, that radius is invalid")

else

    area = 3.142 * (radius ^ 2)

print (area)

endif
```

An **Integrated Development Environment (IDE)** is used to write the program

Identify two features of an **IDE** that might be used when writing the program

.....

..... **(2 marks)**

3 Dru writes the following program using a high-level language.

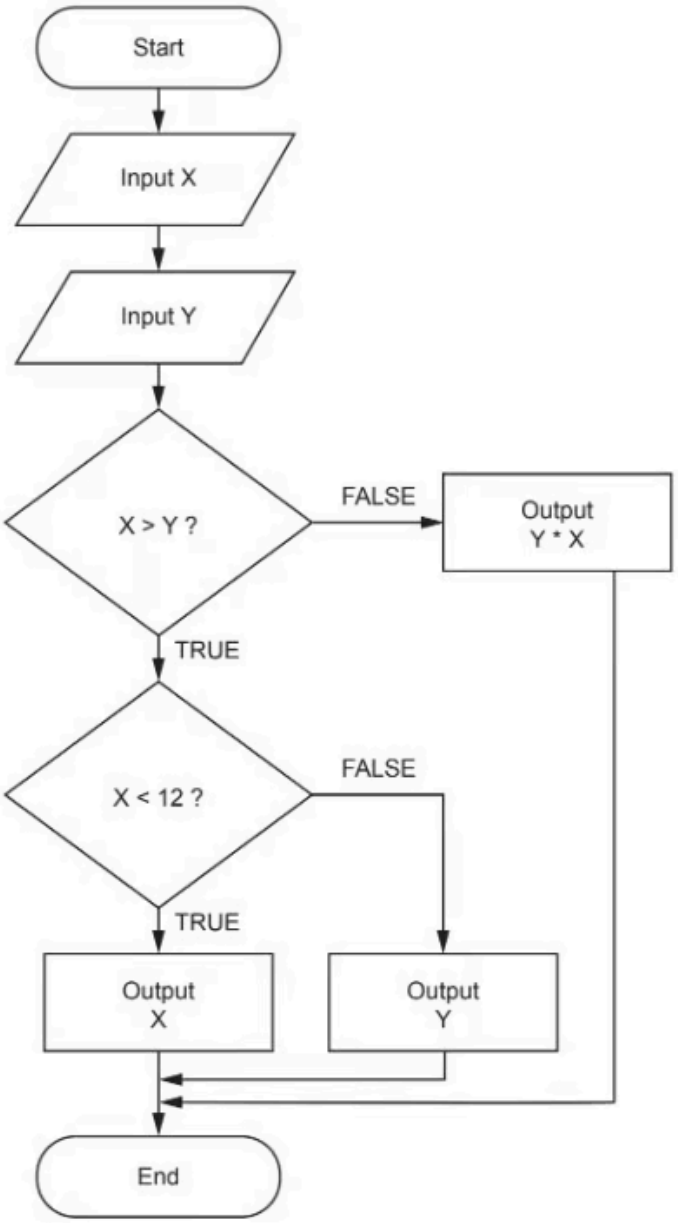
```
function newscore(a,b)
    temp = a*b
    temp = temp + 1
    return temp
endfunction
score = 18
name = "Dru"
print (score)
print ("name")
print (newscore(score,2))
print (score)
```

Describe the advantages of writing the program in a **high-level language** instead of in **assembly language**.

.....

..... **(2 marks)**

4 A programmer creates an algorithm using a flow chart.



The algorithm is written in a high-level language.

The high level code **must be translated into machine code** before a computer processor can execute it.

Describe two methods of **translating high level** code into machine code.

Method 1	
Method 2	

.....

.....

.....

..... **(4 marks)**

5 (a) Victoria is writing a program using a high level language to display the meaning of computer science acronyms that are entered.

The code for her first attempt at this program is shown below.

```
01 a = input("Enter an acronym")

02 if a == "LAN" then

03     print("Local Area Network")

04 elseif a == "WAN" then

05     print("Wide Area Network")

06 ..................................................

07 ..................................................

08 endif
```

Complete the code above to print out an "unknown" message if any other acronym is entered by the user.

.....

..... **(2 marks)**

(b) **Describe** what is meant by a "**high level language**".

.....

..... **(2 marks)**

6 Victoria creates a program using an Integrated **Development Environment (IDE)**.

Describe two tools or **facilities** that an IDE commonly provides.

.....

.....

.....

..... **(4 marks)**

7 A computer game is written in a high-level programming language.

Either a compiler or an interpreter can translate the code.

Describe two differences between how a **compiler** and an **interpreter** would translate the code.

.....

.....

.....

..... **(4 marks)**

8 **Describe** the benefits of using a compiler instead of an interpreter when writing a program.

.....

.....

..... **(3 marks)**

Hard Questions

1 **Explain** the differences between **low-level** and **high-level** programming languages, and provide **one advantage** and **one disadvantage** of each

.....

.....

.....

.....

.....

..... **(6 marks)**

2 **Explain** why high-level languages are considered **portable** and give an example

.....

.....

..... **(3 marks)**

3 **Compare** the **memory usage** implications of using a compiler versus an interpreter

.....

.....

..... **(3 marks)**

© 2026 Save My Exams, Ltd.